

Summary and retrospective

Computational modeling of language
Spring 2026

Final exam

- Mon May 11, 11:30am-2:30pm, Wheeler 102.

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Final exam

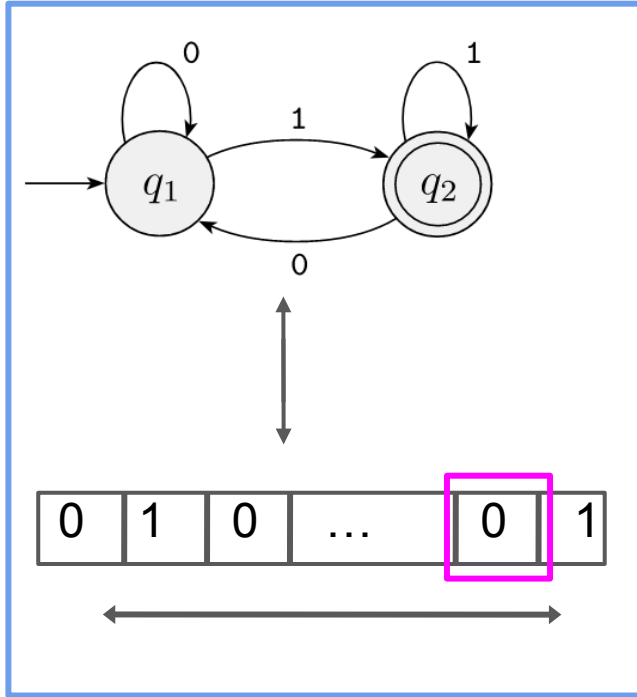
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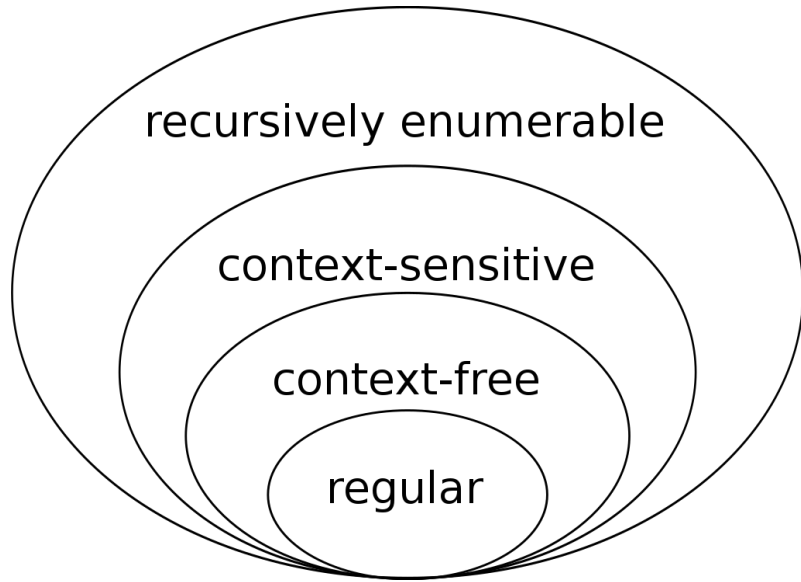
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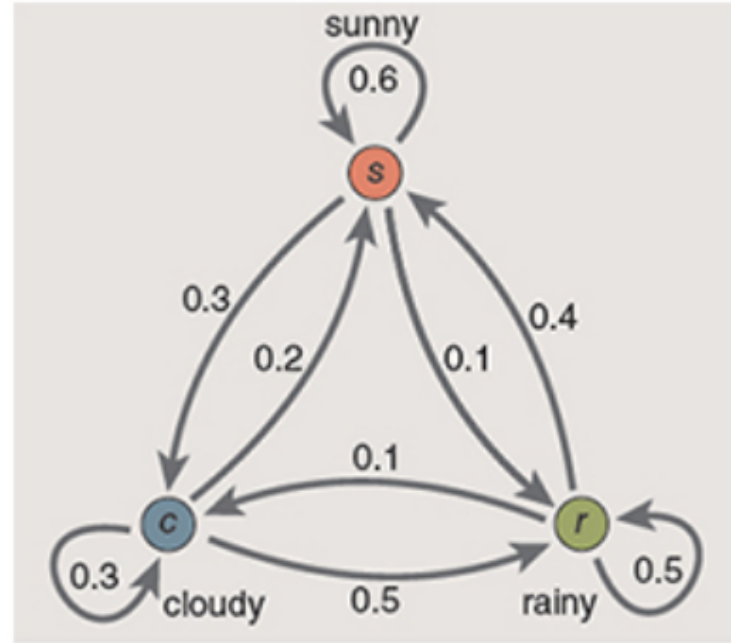
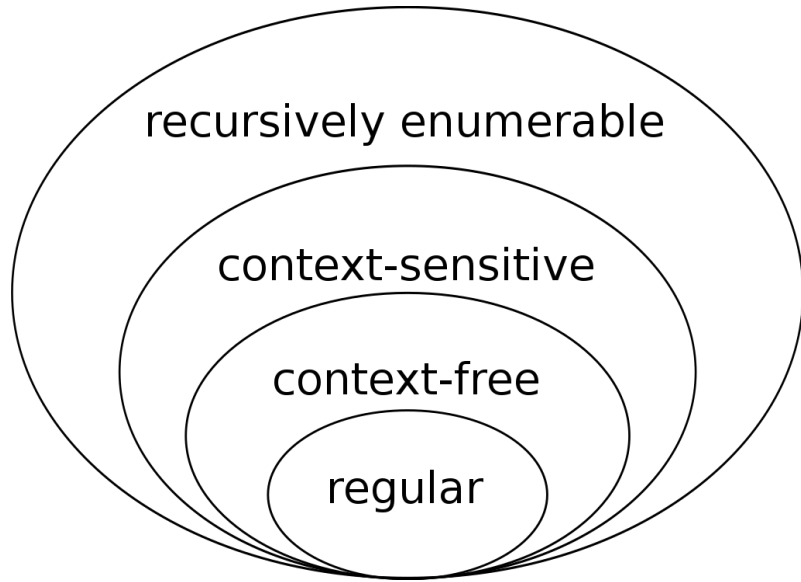
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- Review session during RRR week?

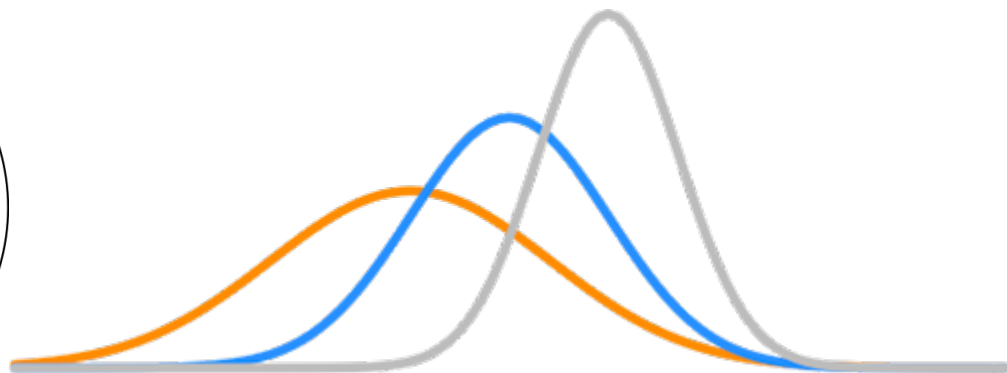
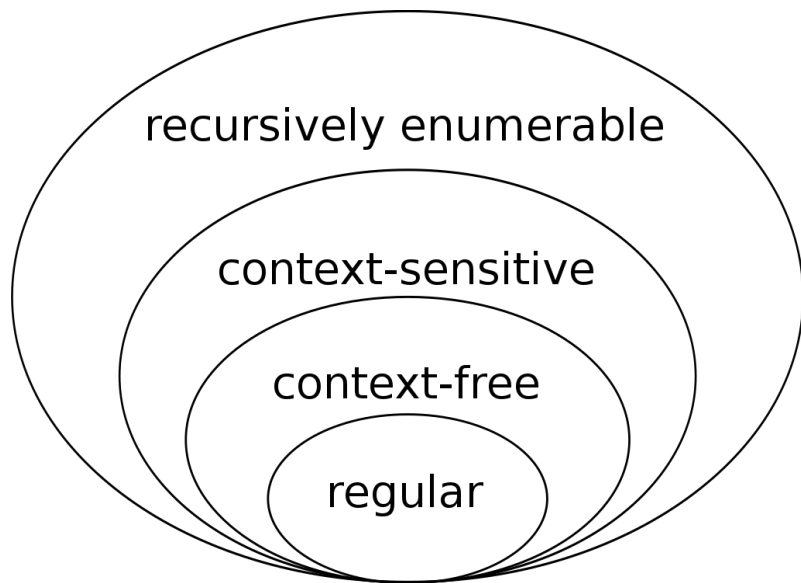


Linguistics and computation
share **deep roots**.





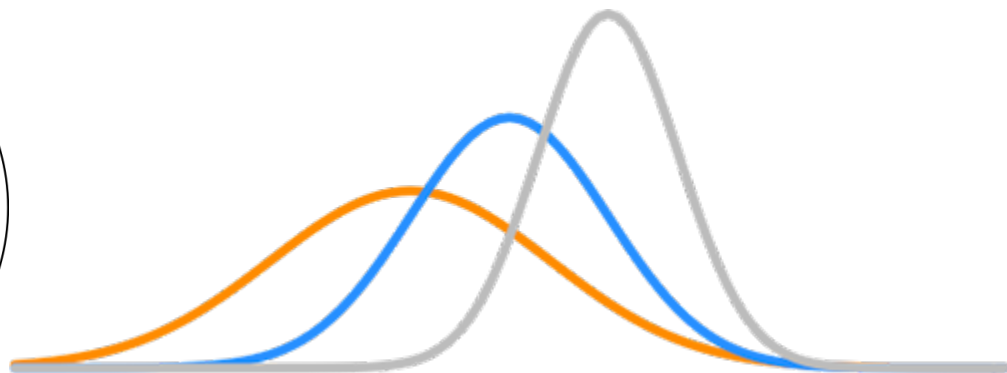
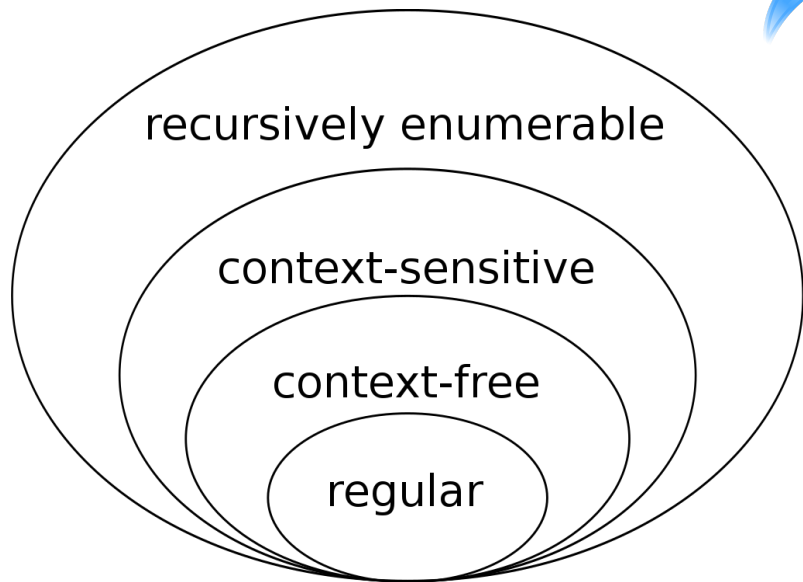




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- Rejection of Markov chains → skepticism toward statistical models **generally**.
- Throwing the **baby** out with the **bathwater**: dismissing an entire class of ideas because of the limitations of one exemplar of that class.



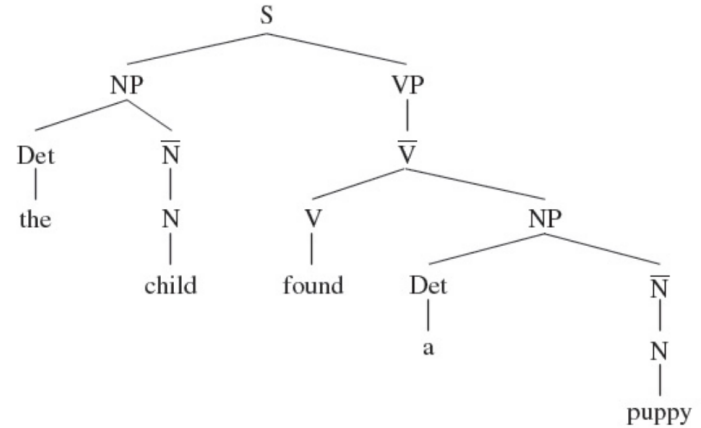
Possible relations of structure and numbers

Arranged in a cline of increasing importance for numbers

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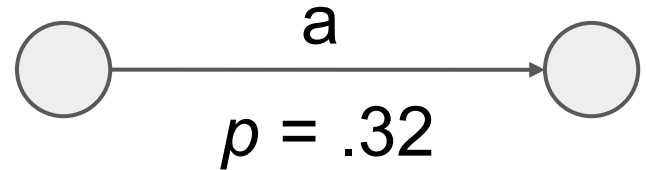
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Possible relations of structure and numbers

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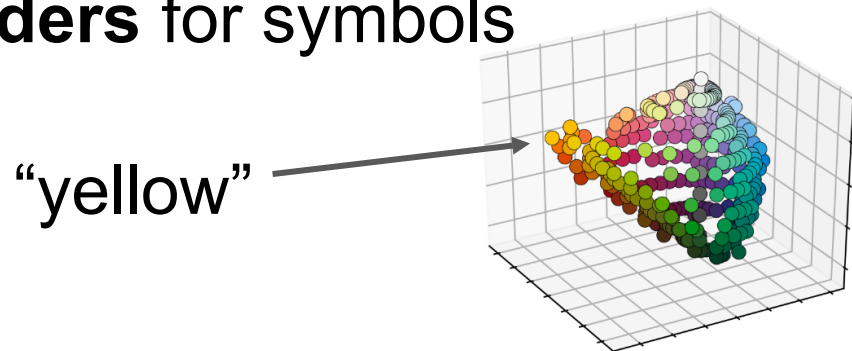
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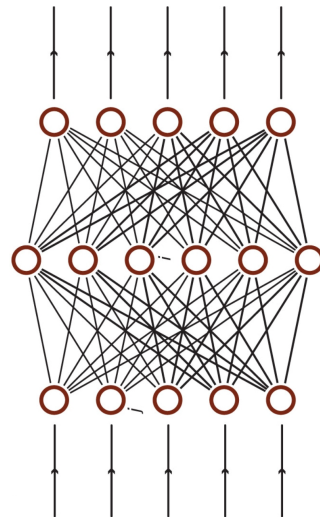
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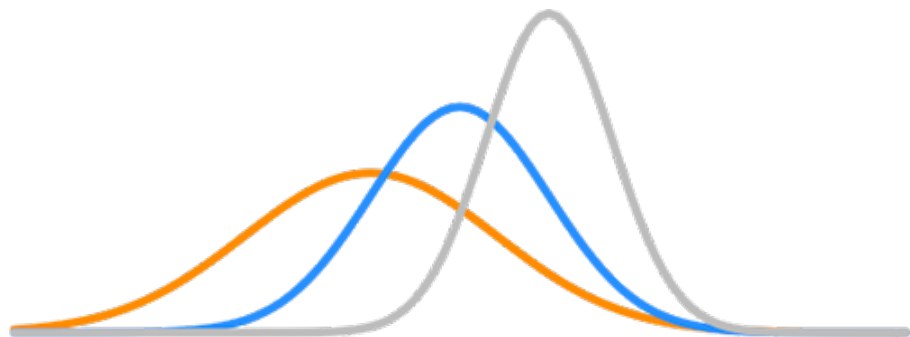
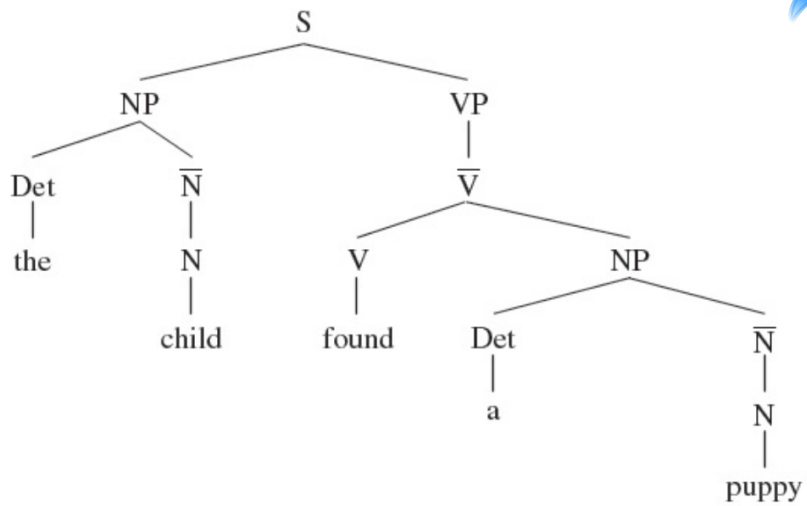


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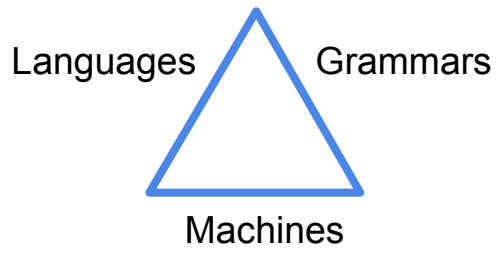
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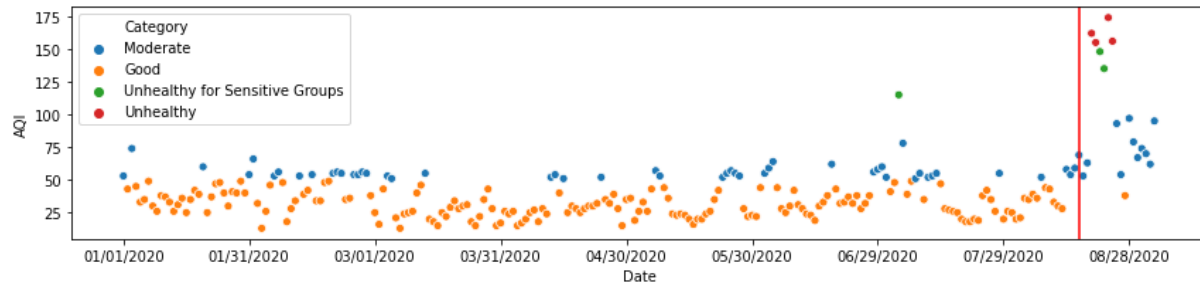
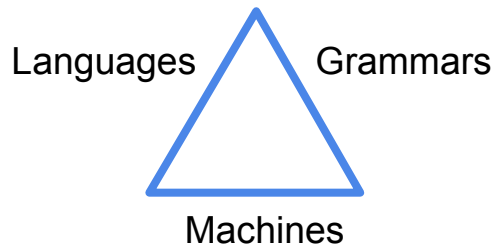


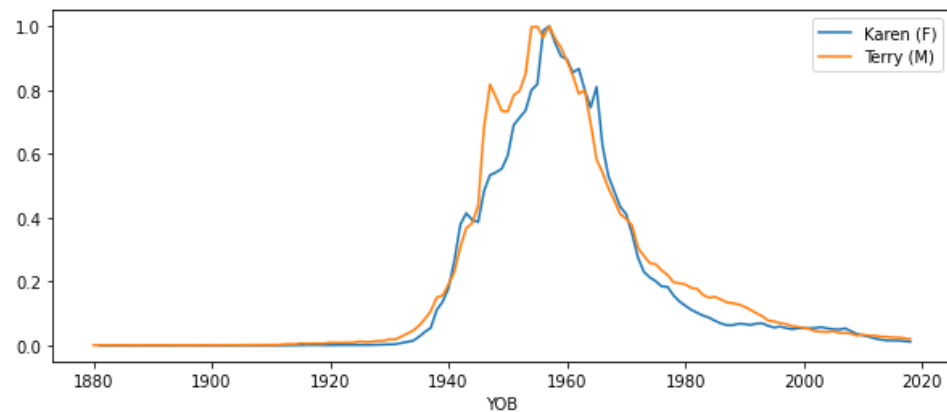
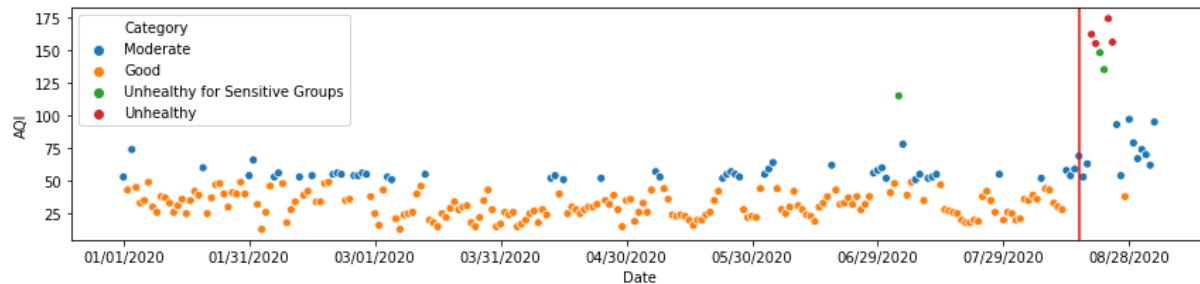
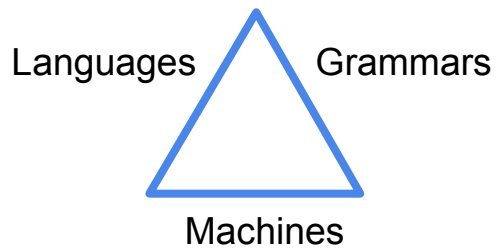


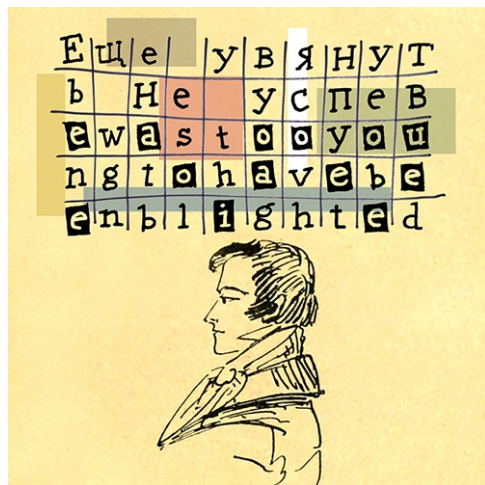
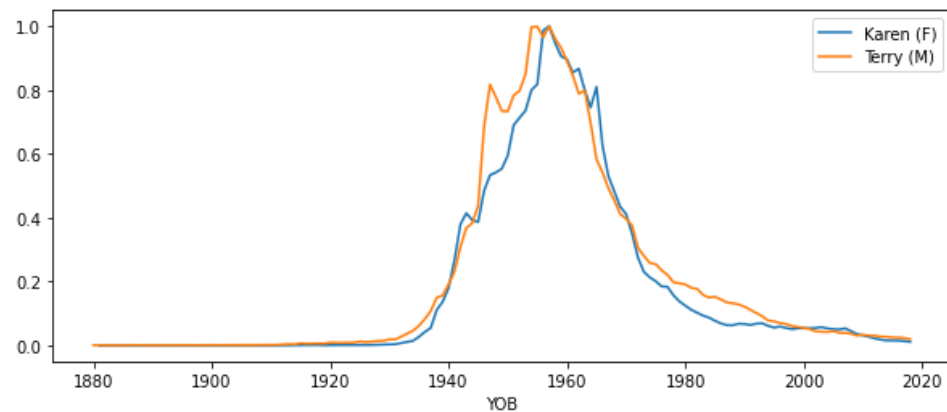
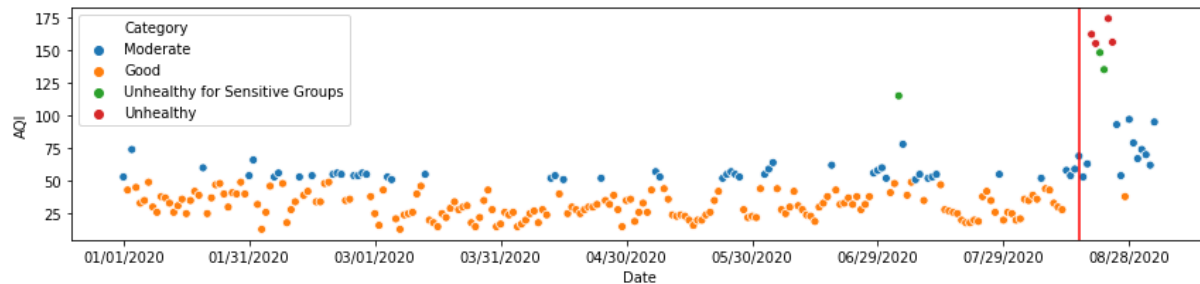
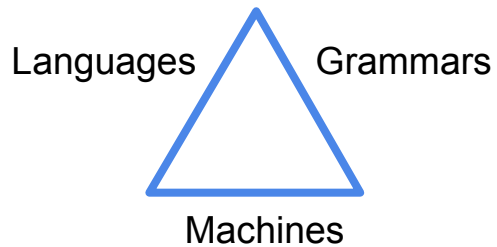
Our class in retrospect

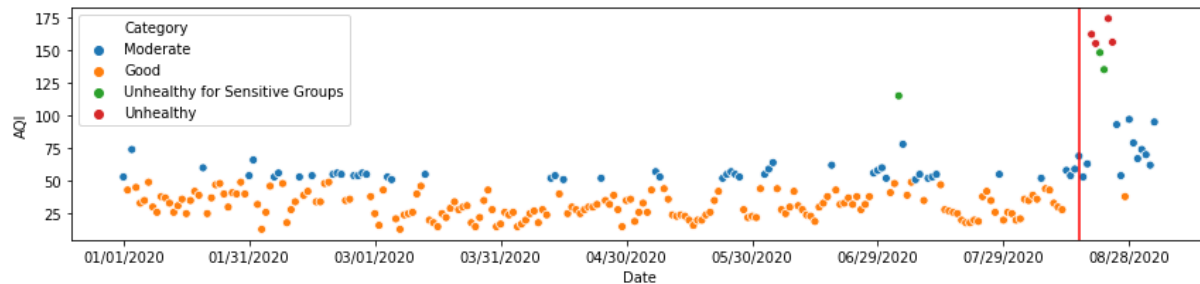
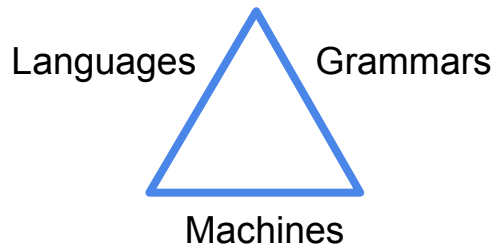
A whirlwind tour



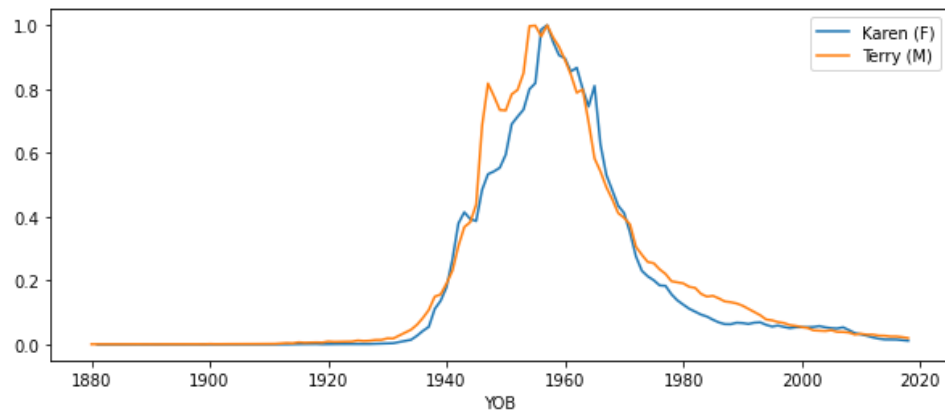
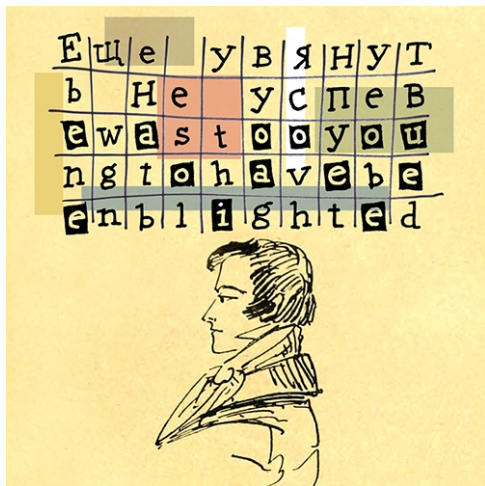


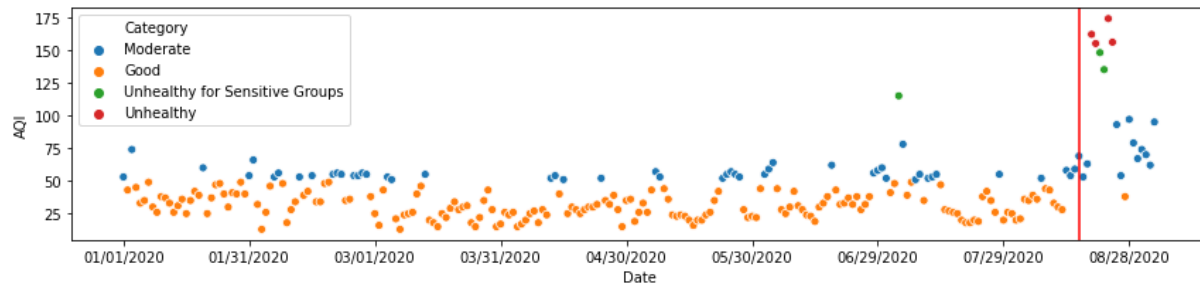
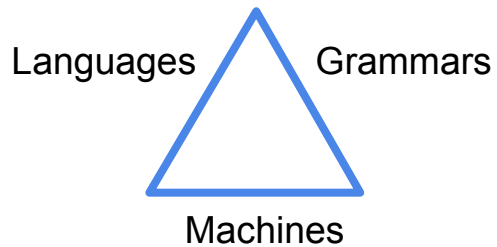




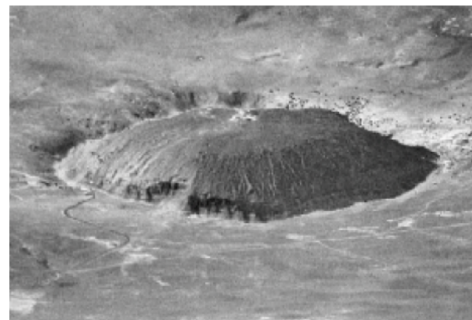
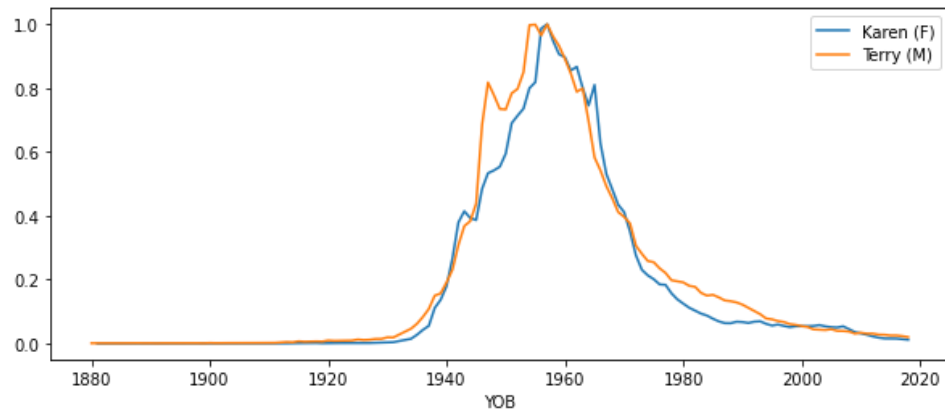
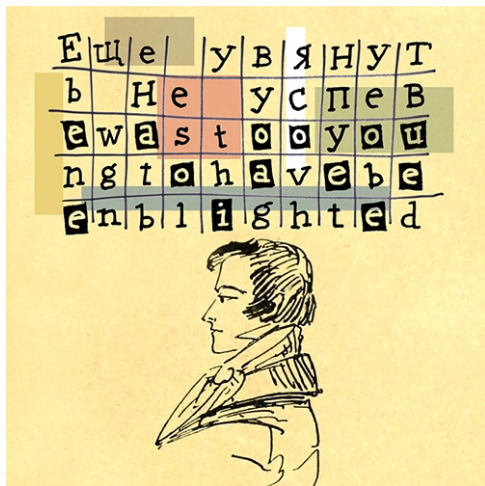


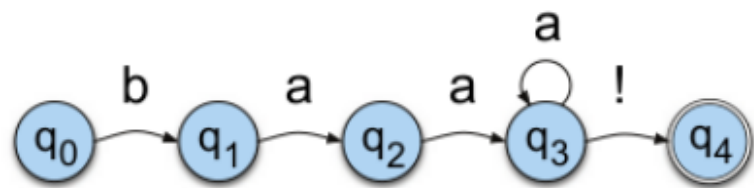
$$p(H|d) \propto p(d|H) p(H)$$

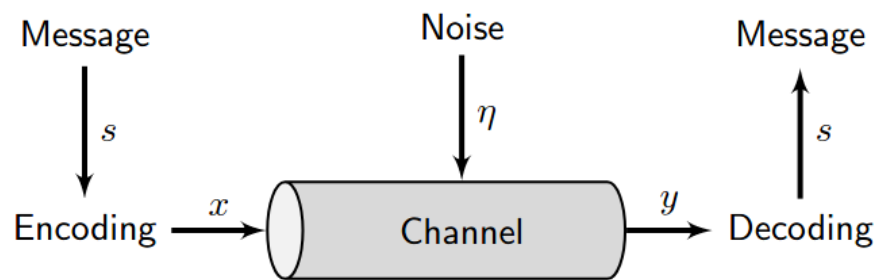
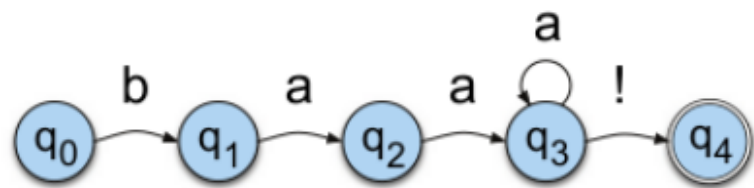


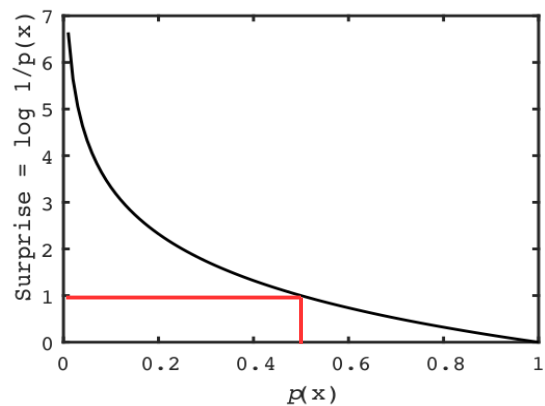
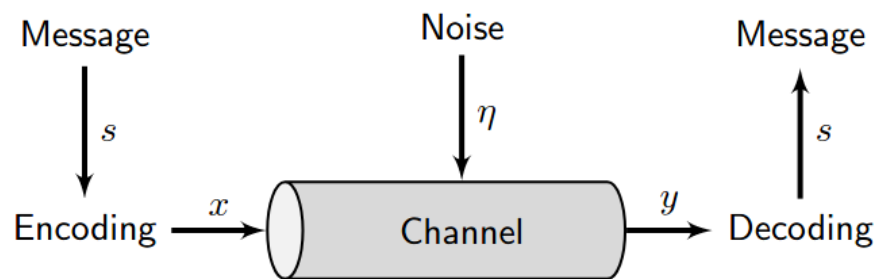
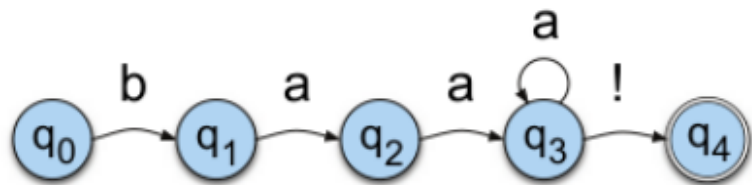


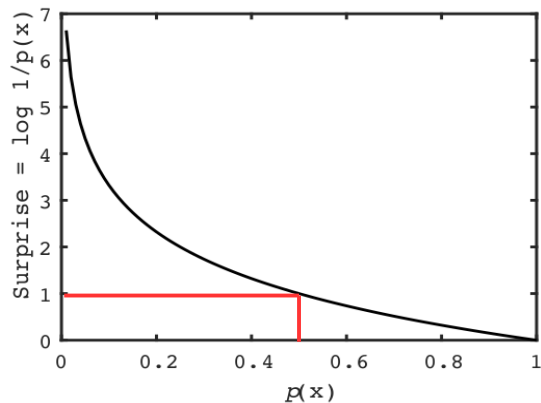
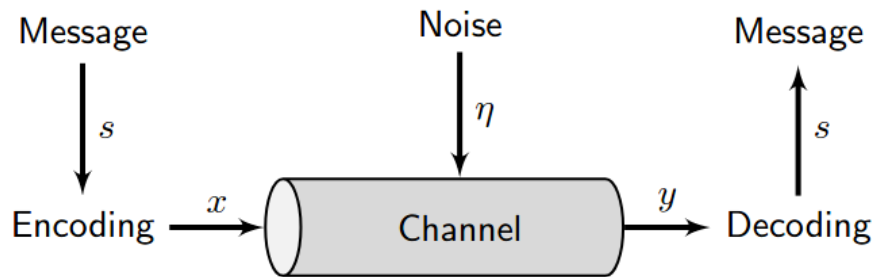
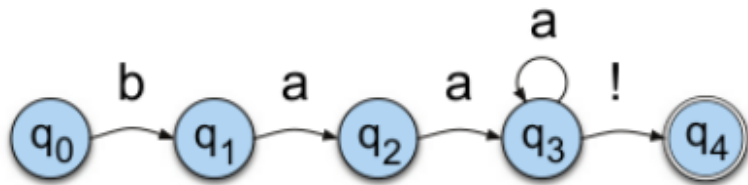
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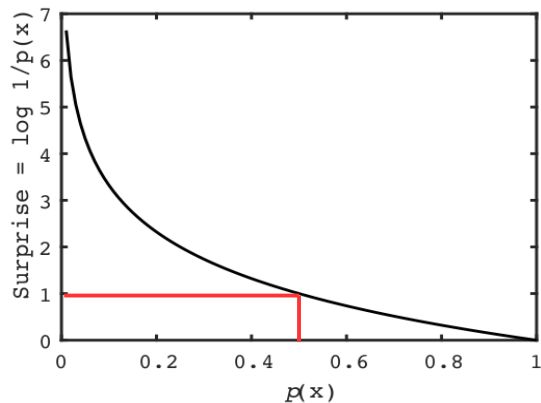
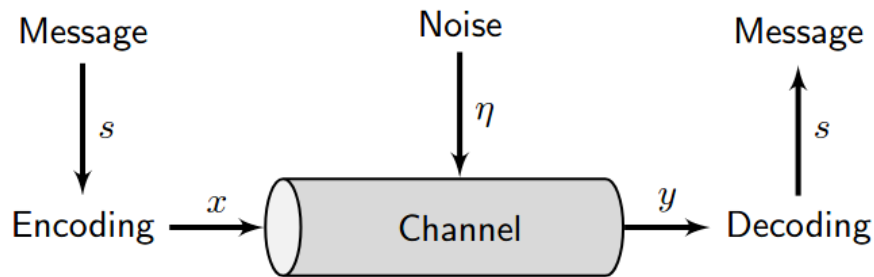
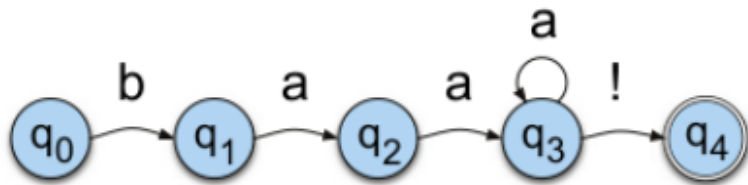




The Zen of Python, by Tim Peters

Beautiful is better than ugly.
 Explicit is better than implicit.
 Simple is better than complex.
 Complex is better than complicated.
 Flat is better than nested.

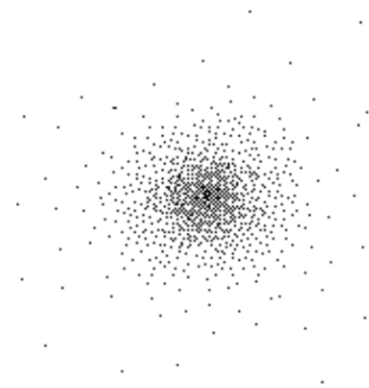
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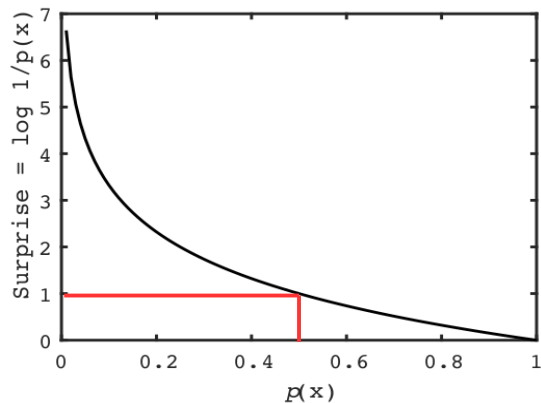
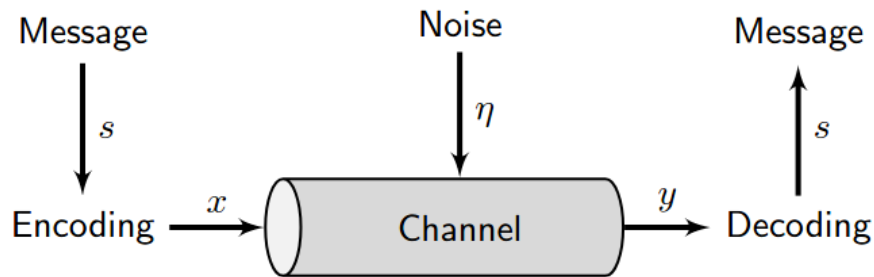
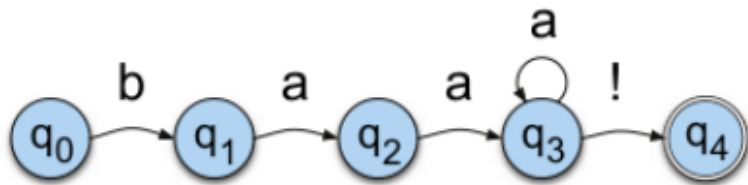


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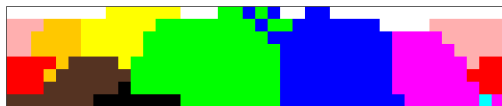
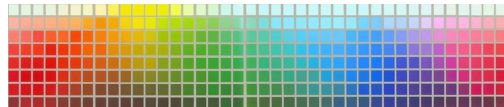
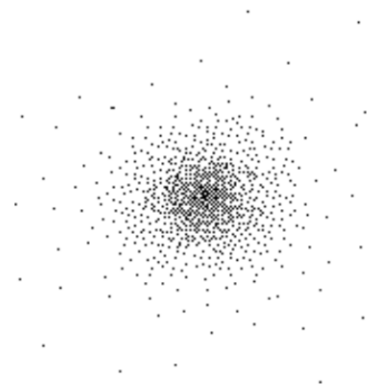


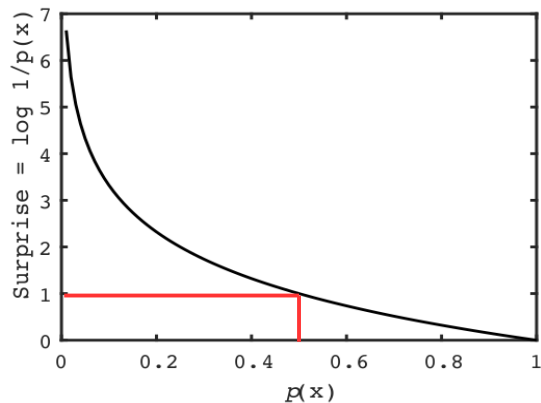
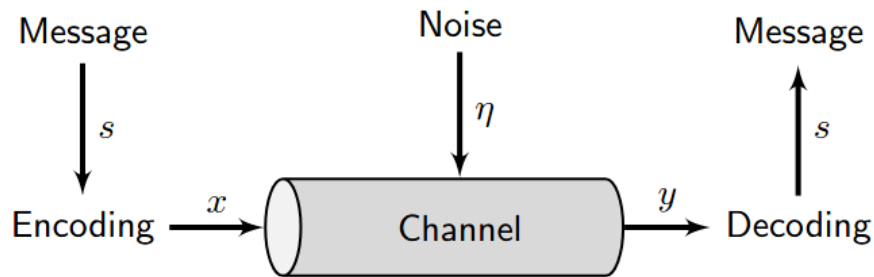
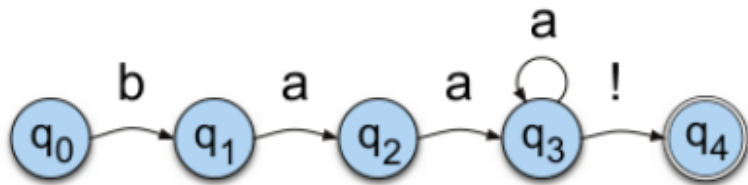


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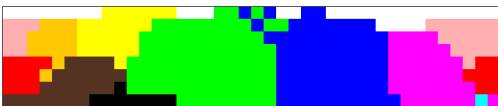
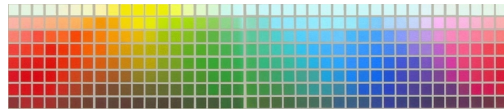
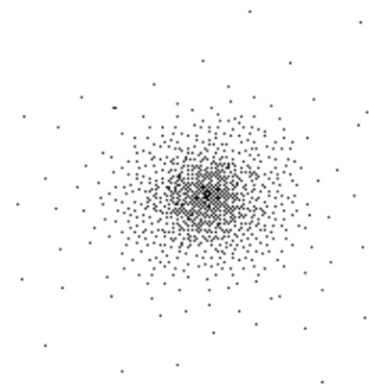




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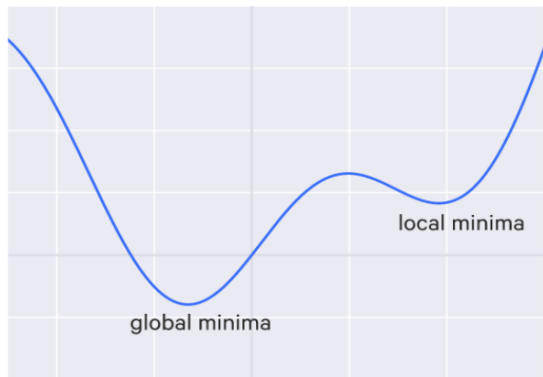
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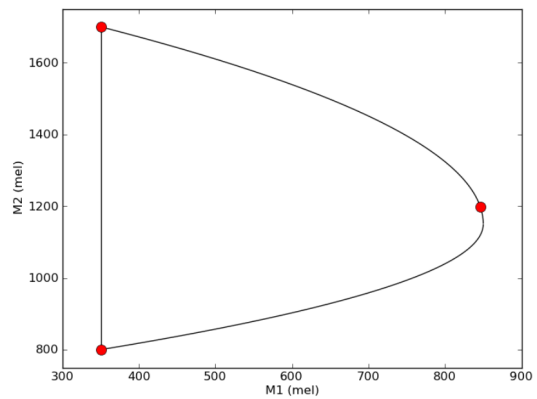
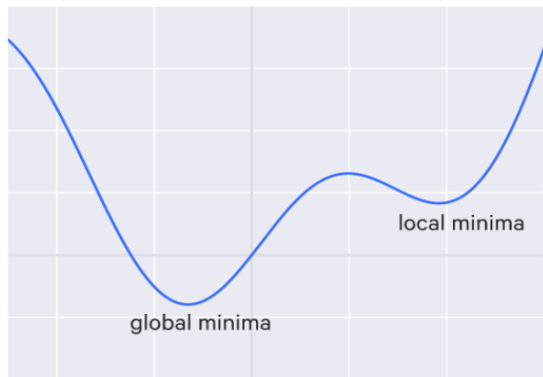


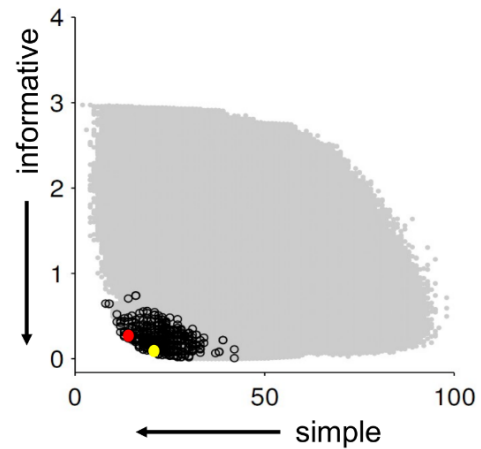
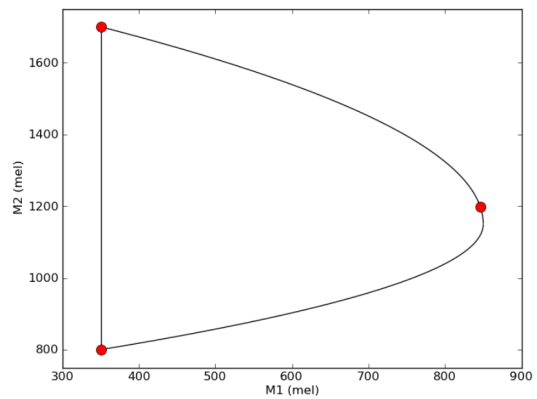
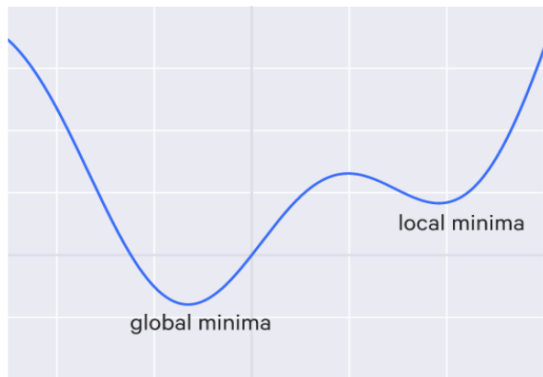
*Computational
theory*

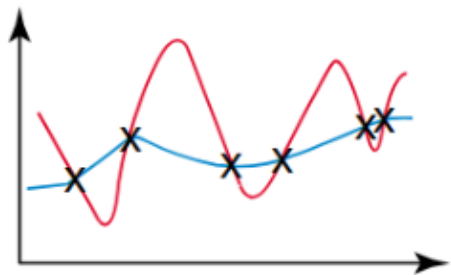
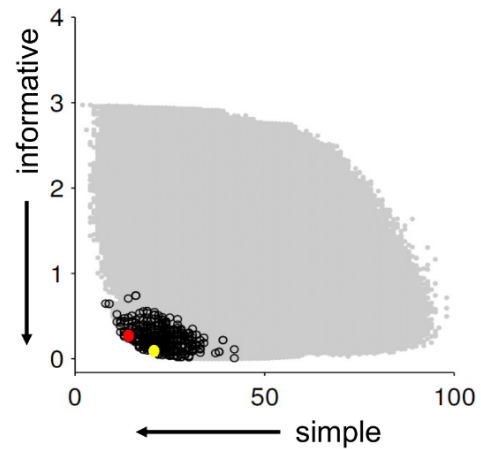
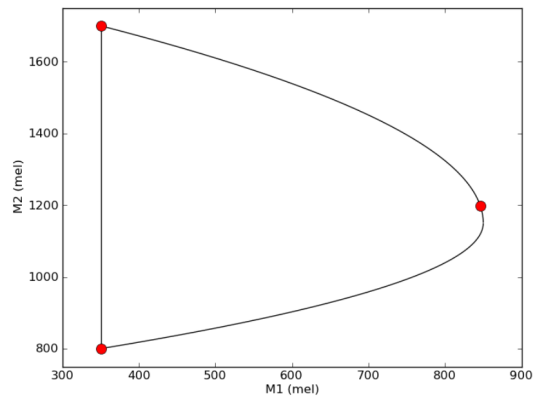
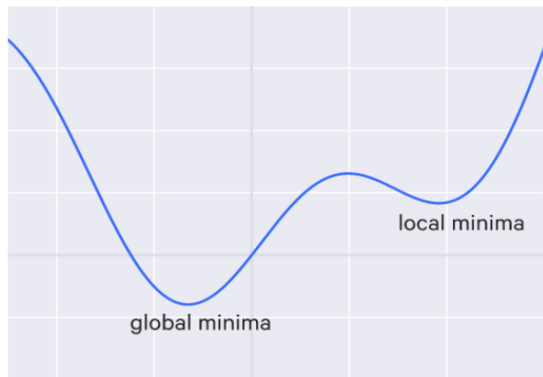
*Representation and
algorithm*

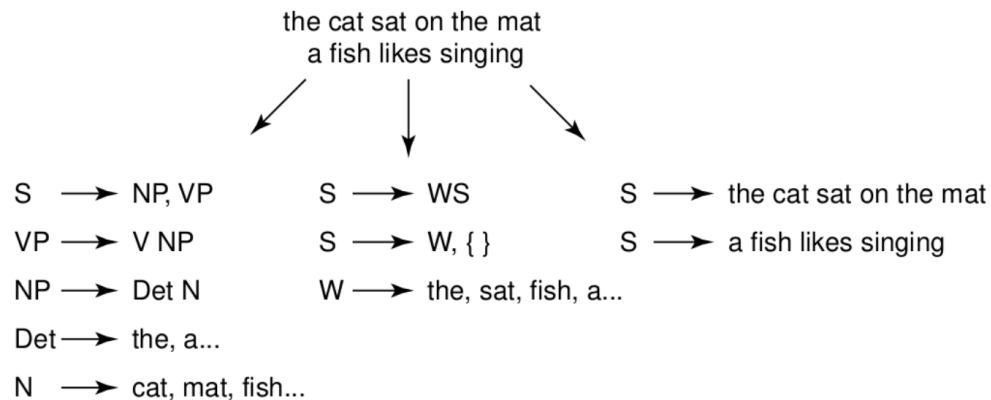
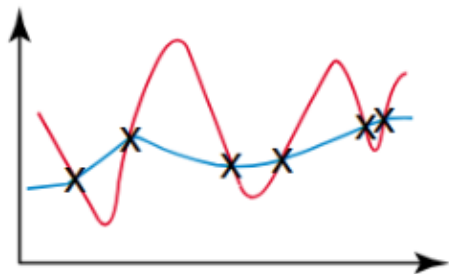
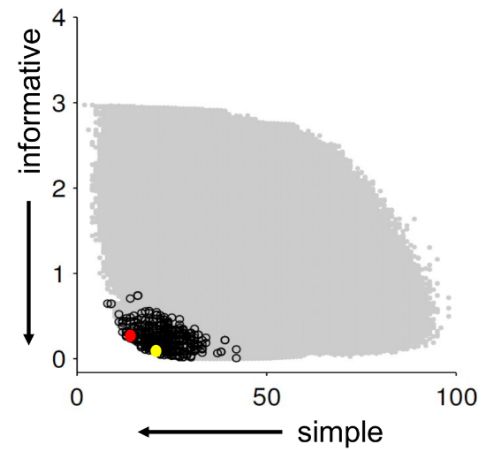
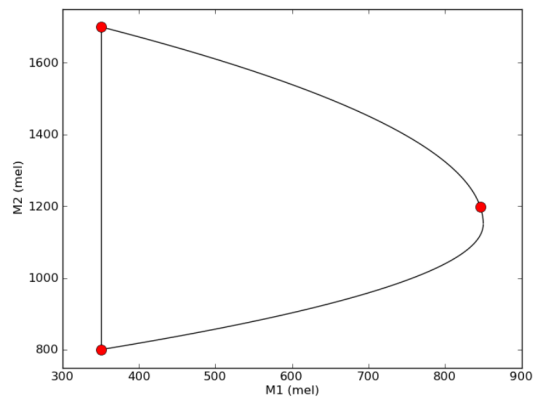
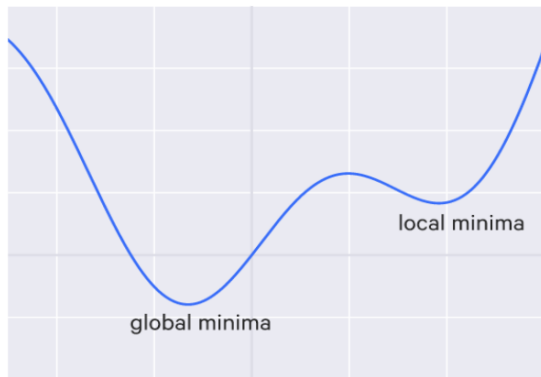
*Hardware
implementation*

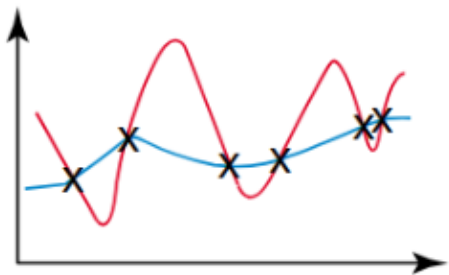
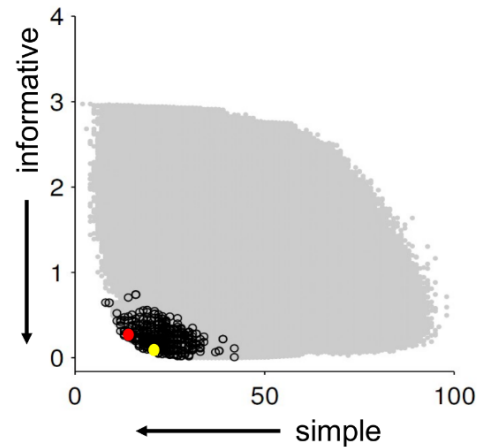
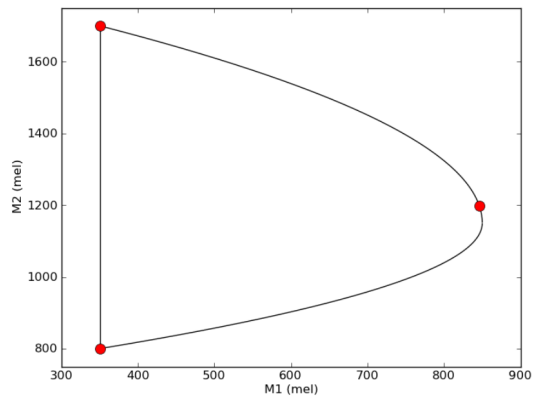
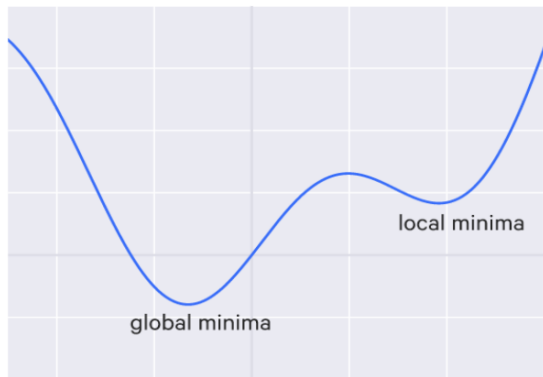




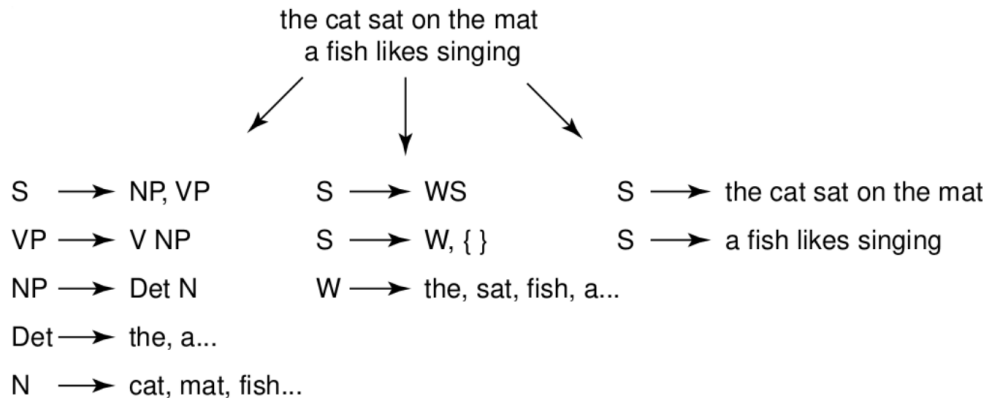








“Wasted” probability mass!



THE SIMPLEST SPAM-FILTER

(used until 2010)

hey	...	1829
I'm	...	1710
no	...	1191
where	...	1012
you	...	985
speak	...	873
learn	...	747
one	...	739

good letters

viagra	...	1552
casino	...	1492
100%	...	1320
credit	...	1184
sale	...	985
press	...	873
free	...	747
enlarge	...	739

spam letters

672 times

«KITTY»

13 times

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

BAYES' THEOREM



NOT SPAM

NAIVE BAYES

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

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FIGURE 2 Languages of South America (two-way Andean core ROC scores)

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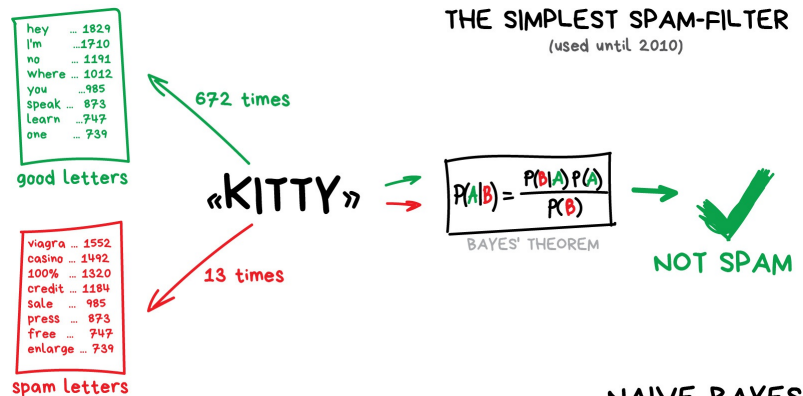
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$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$



FIGURE 2 Languages of South America (two-way Andean core ROC scores)

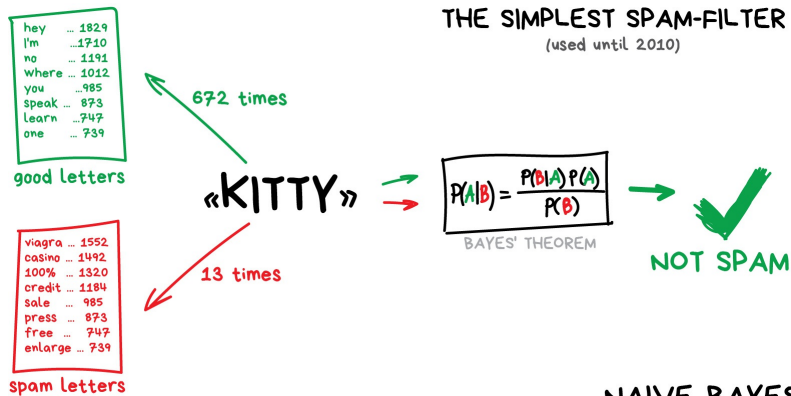


iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
iti					
iti	*!		*	*	
iti	*!		*	*	
iti	*!*			**	

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$



FIGURE 2 Languages of South America (two-way Andean core ROC scores)



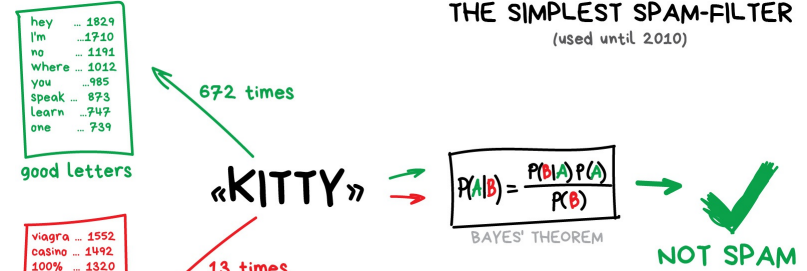
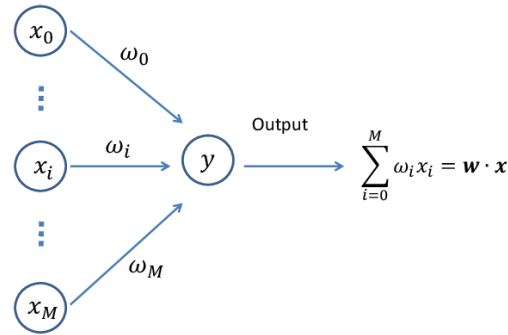
$$\Pr(y|x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^m w_i f_i(y, x)\right),$$

iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
iti					
rti	*!		*	*	
itr	*!		*	*	
rti	*!*			**	

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$



FIGURE 2 Languages of South America (two-way Andean core ROC scores)



NAIVE BAYES

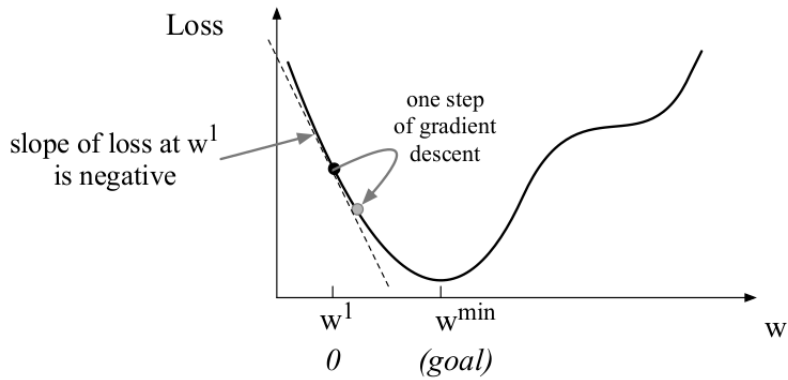
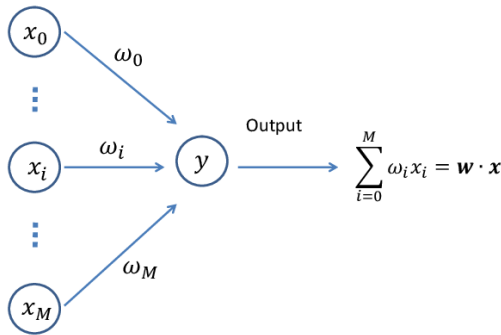
$$\Pr(y|x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^m w_i f_i(y, x)\right),$$

iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
iti					
r <i>t</i> i	*!		*	*	
i <i>t</i> i	*!		*	*	
r <i>t</i> i	*!*			**	

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$



FIGURE 2 Languages of South America (two-way Andean core ROC scores)



hey ... 1929
I'm ... 1710
no ... 1191
where ... 1012
you ... 985
speak ... 873
learn ... 747
one ... 739

good letters

viagra ... 1552
casino ... 1492
100% ... 1320
credit ... 1184
sale ... 985
press ... 873
free ... 747
enlarge ... 739

spam letters

672 times

«KITTY»

13 times

THE SIMPLEST SPAM-FILTER
(used until 2010)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

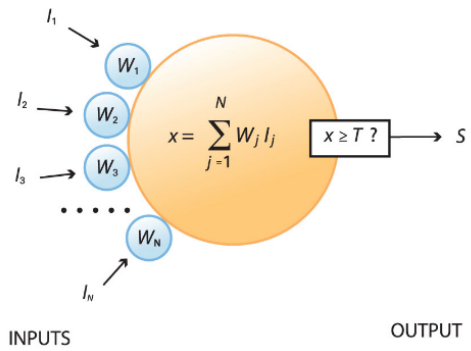
BAYES' THEOREM

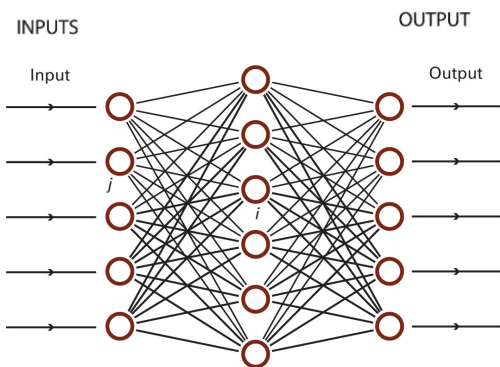
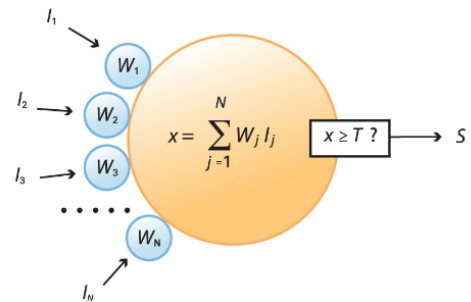
✓
NOT SPAM

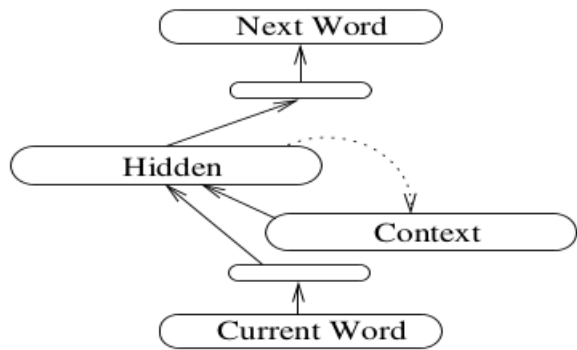
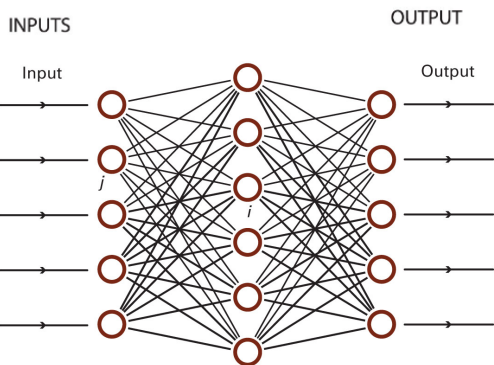
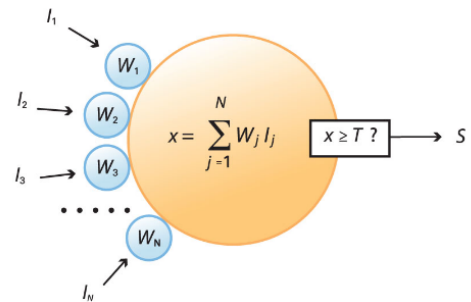
NAIVE BAYES

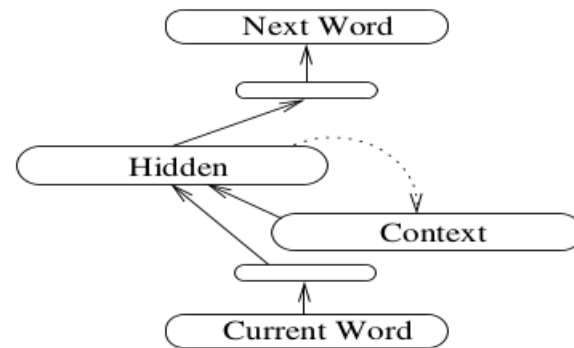
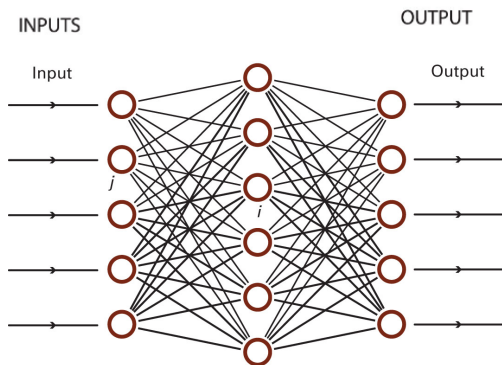
$$\Pr(y|x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^m w_i f_i(y, x)\right),$$

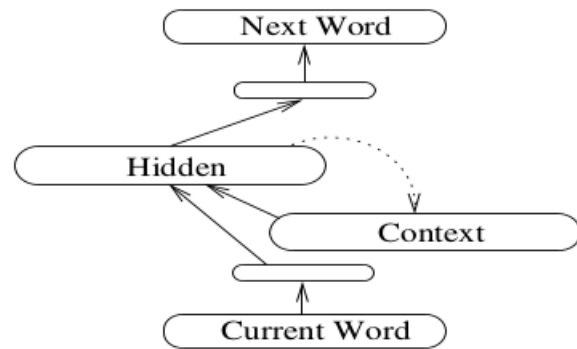
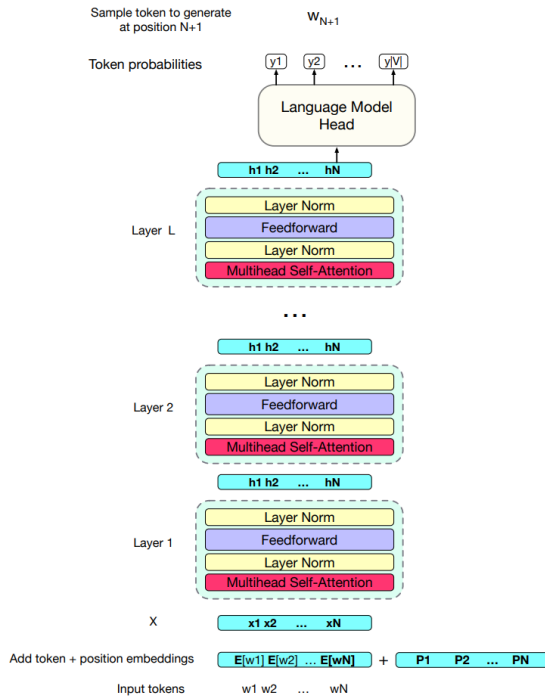
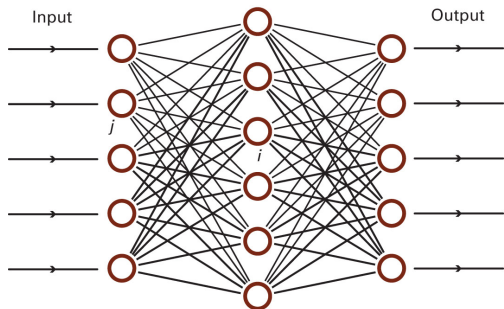
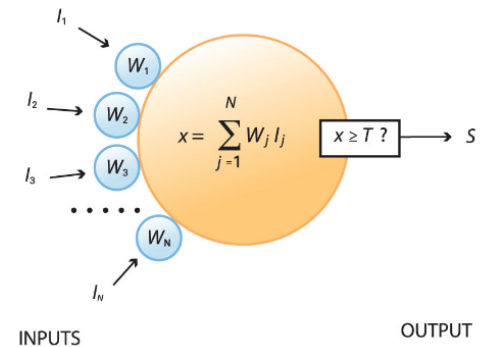
iti	*[rtr / hi]	PARSE (rtr)	*GESTURE (contour)	PARSE (atr)	*[atr / lo]
iti					
rti	*!		*	*	
iti	*!		*	*	
rti	*!*			**	



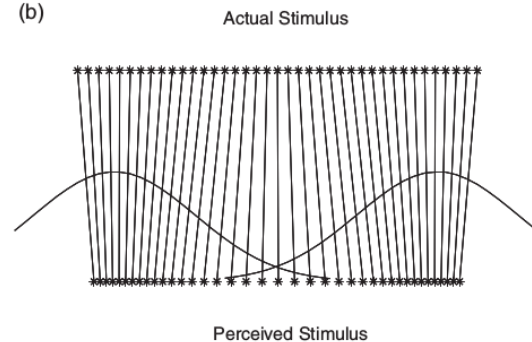


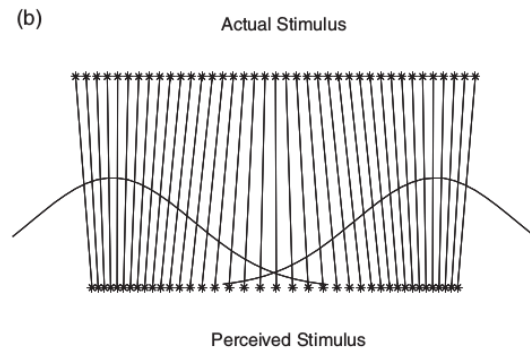
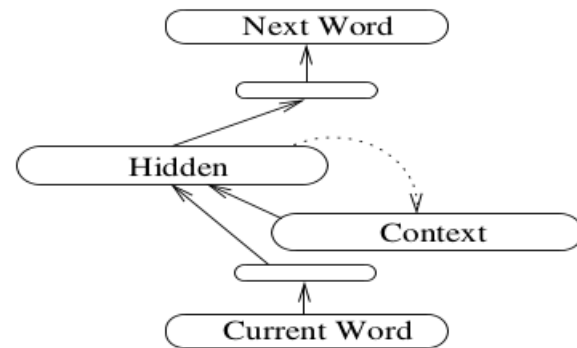
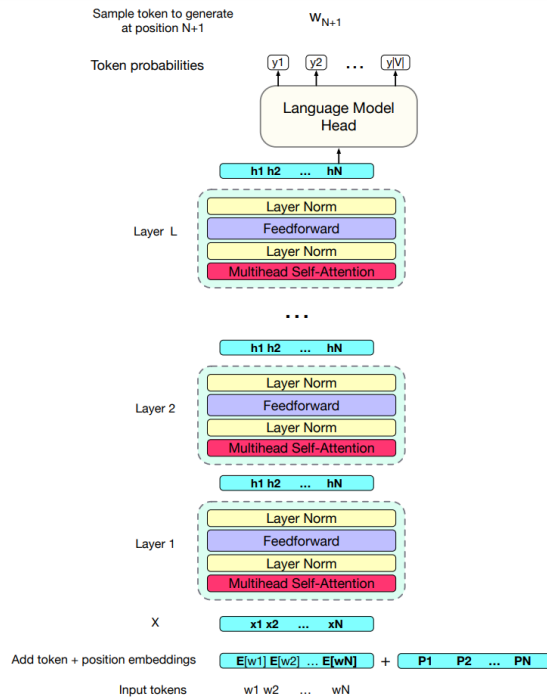
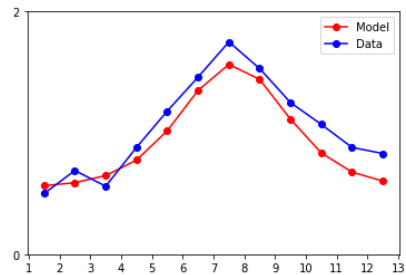
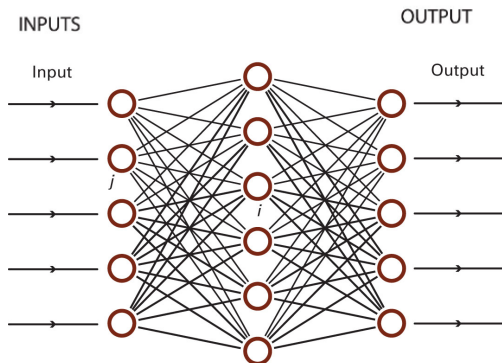
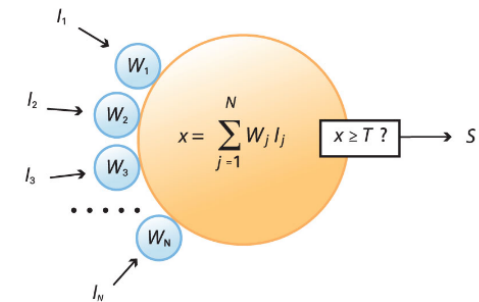


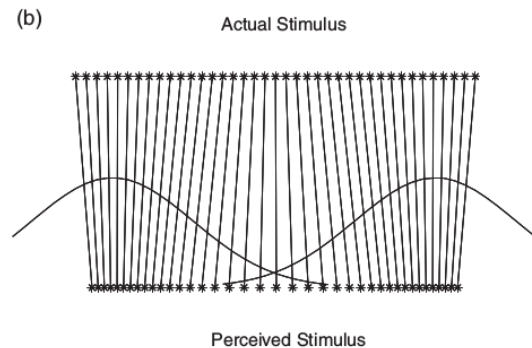
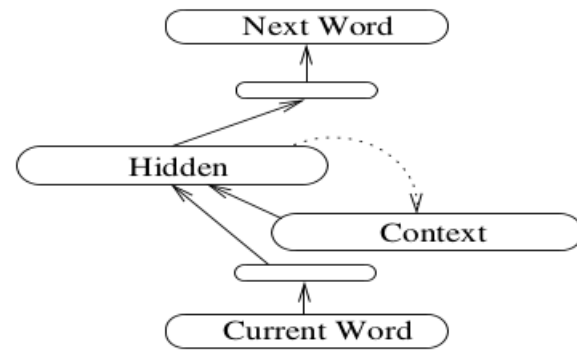
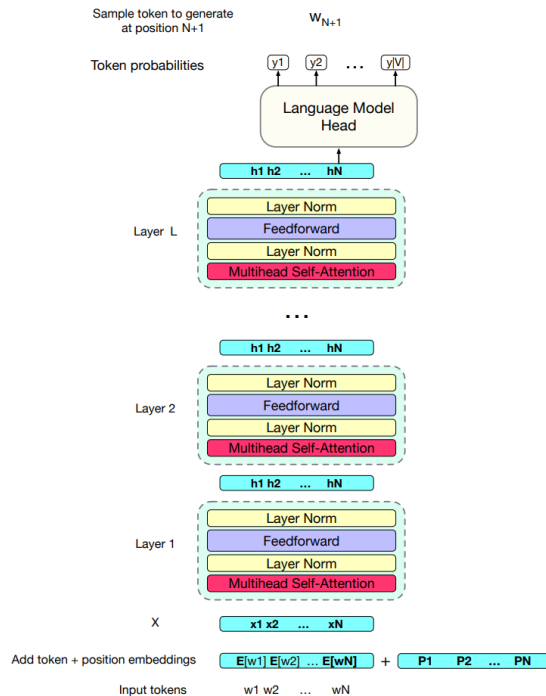
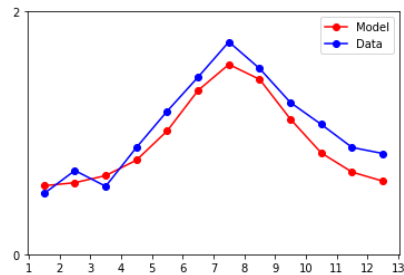
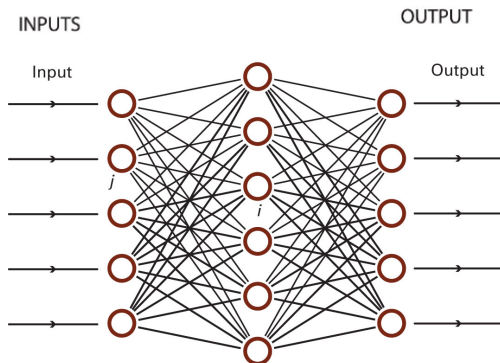
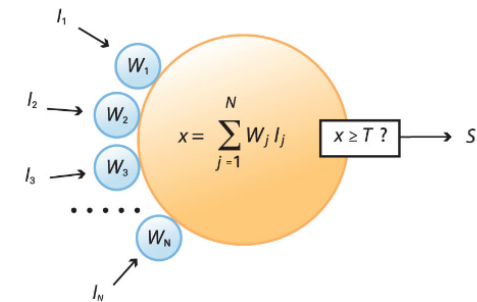




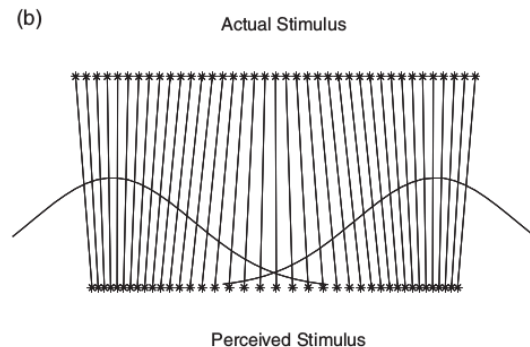
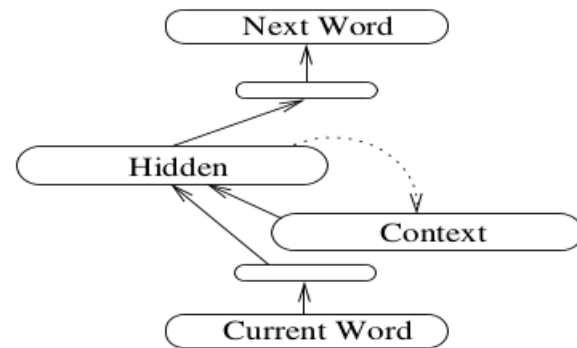
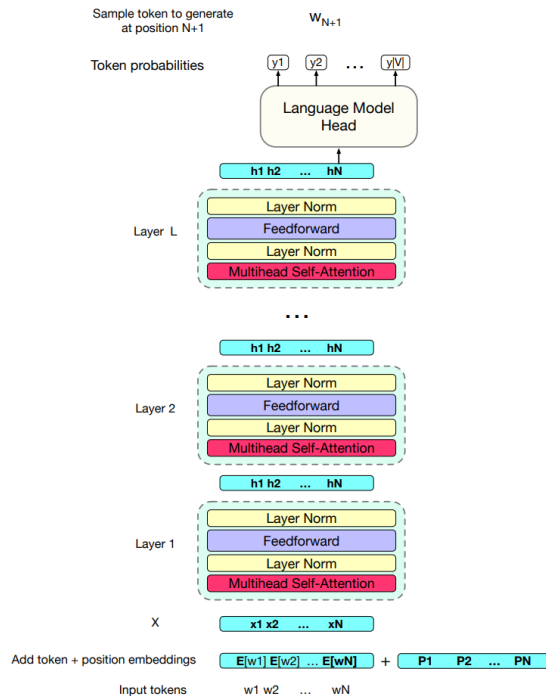
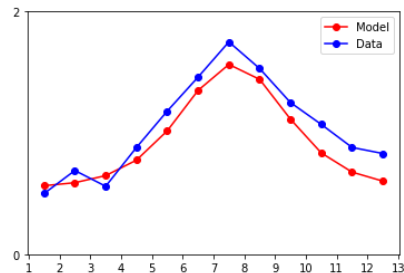
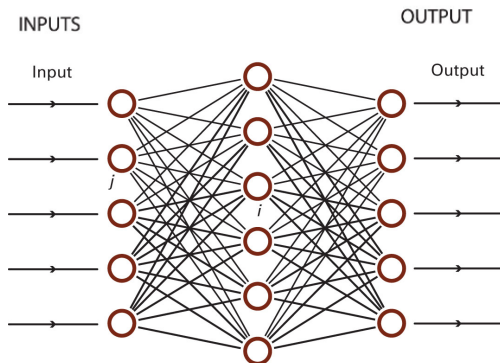
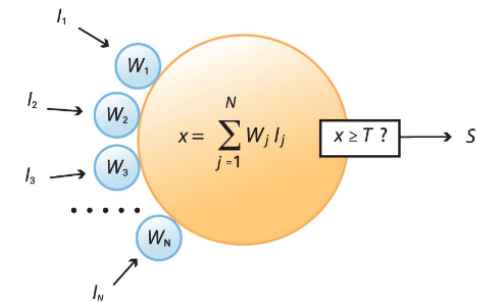
(b)







$$E[T|S] = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_S^2} S + \frac{\sigma_S^2}{\sigma_c^2 + \sigma_S^2} \sum_c p(c|S) \mu_c$$



$$E[T|S] = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_S^2} S + \frac{\sigma_S^2}{\sigma_c^2 + \sigma_S^2} \sum_c p(c|S) \mu_c$$

Thanks to my colleagues for suggesting readings!

- Lev: Andean areality via naive Bayes
- Sharon and Gasper: Maxent OT
- Keith: Categorical perception of speech

Your thoughts?

- Take a look back over our bCourses site and/or syllabus.
- If you could identify **one idea to definitely keep** for the next iteration of this class – and **one to maybe not keep** – what would those be?

Some things we haven't discussed

- Current uses of formal language theory
- Agent-based simulation (only briefly)
- Distributional semantics
- Probabilistic graphical models
- ...

What next?

- It depends on what you're looking for!
- Let me know your interests, and I'm happy to make suggestions.

Have a restful and enjoyable summer!

And I'm curious to hear your opinions,
questions, comments, etc. - stay in touch!

Course evaluations

Please take a few minutes to fill out your
course evaluation!